Question

1×10^{-3} mol of $Cd(CN)_2$ is placed in $1\ L$ of buffer solution at $pH = 8.5$. It is known that the cumulative stability constants of CN^- to Cd^{2+} are $10^{5.48}, 10^{10.60}, 10^{15.23}, 10^{18.78}$, respectively, and the solubility product constant of $Cd(OH)_2$ is $K_{sp} = 4 \times 10^{-13}$, and the dissociation constant of HCN is $K_a = 6.2 \times 10^{-10}$. Coordination of cadmium species and OH^- is ignored.

The following statements are made:

1. The species conservation of cyanide ions in the solution is $[CN^-] + [HCN] + [Cd(CN)^+] + 2[Cd(CN)_2] + 3[Cd(CN)_3^-] + 4[Cd(CN)_4^{2-}] = 2 \times 10^{-3}\ mol/L$

2. The charge conservation in the solution is $2[Cd^{2+}] + [Cd(CN)^+] + [H^+] = [Cd(CN)_3^-] + 2[Cd(CN)_4^{2-}] + [CN^-] + [OH^-]$

3. Let $[CN^-] = a.b \times 10^{-c}\ mol/L$ (retain two significant figures), then $a + b + c = 10$

4. The system will produce $Cd(OH)_2$ precipitate

What is the sum of the serial numbers of all correct statements?

A. 1

B. 3

C. 4

D. 6

E. 7

F. 8

G. All of the above options are incorrect.

Answer

Correct Answer: C

Detailed Explanation

This question cannot be solved using charge conservation because the buffer ion concentration changes before and after the addition of $Cd(CN)_2$, its relative buffer ion concentration is small, but the relative concentrations of cadmium-containing and cyanide-containing species are large and unknown. Statement 1 regarding material balance is correct, and statement 2 regarding charge conservation is incorrect because buffer ions are not considered.

CHECKPOINT 1 PTS

Statement 1 is correct

CHECKPOINT 2 PTS

Statement 2 is incorrect

According to material balance $[CN^-] + [HCN] + [Cd(CN)^+] + 2[Cd(CN)_2] + 3[Cd(CN)_3^-] + 4[Cd(CN)_4^{2-}] = 2 \times 10^{-3}\ mol/L$, express the concentrations of species other than CN^- in terms of $[CN^-]$.

$$[HCN] = \frac{[CN^-][H^+]}{K_a}$$

CHECKPOINT 0.5 PTS

$$[\mathrm{HCN}] = \frac{[\mathrm{CN}^-][\mathrm{H}^+]}{K_\mathrm{a}}$$

According to the distribution coefficient, we have:

$$[\mathrm{Cd(CN)}^+] = C_\mathrm{Cd} \cdot \frac{\beta_1[\mathrm{CN}^-]}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

$$[\mathrm{Cd(CN)}_2] = C_\mathrm{Cd} \cdot \frac{\beta_2[\mathrm{CN}^-]^2}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

$$[\mathrm{Cd(CN)}_3^-] = C_\mathrm{Cd} \cdot \frac{\beta_3[\mathrm{CN}^-]^3}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

$$[\mathrm{Cd(CN)}_4^{2-}] = C_\mathrm{Cd} \cdot \frac{\beta_4[\mathrm{CN}^-]^4}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

Where $C_\mathrm{Cd} = 1 \times 10^{-3}\ \mathrm{mol/L}$.

CHECKPOINT 0.5 PTS

$$[\mathrm{Cd(CN)}^+] = C_\mathrm{Cd} \cdot \frac{\beta_1[\mathrm{CN}^-]}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

CHECKPOINT 0.5 PTS

$$[\mathrm{Cd(CN)}_2] = C_\mathrm{Cd} \cdot \frac{\beta_2[\mathrm{CN}^-]^2}{1+\beta_1[\mathrm{CN}^-]+\beta_2[\mathrm{CN}^-]^2+\beta_3[\mathrm{CN}^-]^3+\beta_4[\mathrm{CN}^-]^4}$$

CHECKPOINT 0.5 PTS

$$[\mathrm{Cd(CN)_3^-}] = C_{\mathrm{Cd}} \cdot \frac{\beta_3[\mathrm{CN^-}]^3}{1+\beta_1[\mathrm{CN^-}]+\beta_2[\mathrm{CN^-}]^2+\beta_3[\mathrm{CN^-}]^3+\beta_4[\mathrm{CN^-}]^4}$$

CHECKPOINT 0.5 PTS

$$[\mathrm{Cd(CN)_4^{2-}}] = C_{\mathrm{Cd}} \cdot \frac{\beta_4[\mathrm{CN^-}]^4}{1+\beta_1[\mathrm{CN^-}]+\beta_2[\mathrm{CN^-}]^2+\beta_3[\mathrm{CN^-}]^3+\beta_4[\mathrm{CN^-}]^4}$$

Substituting into the material balance equation, we obtain $[\mathrm{CN^-}] = 1.4 \times 10^{-5}\ \mathrm{mol/L}$. According to the notation in statement 3, $a+b+c=10$, so statement 3 is correct.

CHECKPOINT 0.5 PTS

$$[\mathrm{CN^-}] = 1.4 \times 10^{-5}\ \mathrm{mol/L}$$

CHECKPOINT 0.5 PTS

$$a+b+c=10$$

According to the distribution coefficient, $[\mathrm{Cd^{2+}}] = C_{\mathrm{Cd}} \cdot \frac{1}{1+\beta_1[\mathrm{CN^-}]+\beta_2[\mathrm{CN^-}]^2+\beta_3[\mathrm{CN^-}]^3+\beta_4[\mathrm{CN^-}]^4} = 5.9 \times 10^{-5}\ \mathrm{mol/L}$

$$Q = [\mathrm{Cd^{2+}}][\mathrm{OH^-}]^2/(1\ \mathrm{mol/L})^3 = 5.9 \times 10^{-16} < K_{\mathrm{sp}}$$

CHECKPOINT 1 PTS

$Q = 5.9 \times 10^{-16} < K_{sp}$

$Cd(OH)_2$ precipitate will not be produced, so statement 4 is incorrect.

In summary, the correct statements are 1 and 3, so choose option C.